CS502 - Fundamentals of Algorithms (Quiz No. 4)

Question # 4 of 10 (Start time: 02:36:01 PM, 02 July 2025)

Breadth-first search is a traversal strategy for_____.

Select the correct option

○	Tree
○	Link List
●	Graph
○	Array

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Question # 9 of 10 (Start time: 02:37:45 PM, 02 July 2025)

If you find yourself in maze the better traversel approach will be :

Select the correct option

○	Level order
●	DFS
○	BFS
○	BFS and DFS both are valid

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Question # 8 of 10 (Start time: 02:37:33 PM, 02 July 2025)

A graph is said to be acyclic if it contains _____

Select the correct option

○	exactly one cycle
○	at least one cycle
●	no cycles
○	always more than one cycle

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Question # 1 of 10 (Start time: 02:35:23 PM, 02 July 2025)

The ancestor and descendent relation can be nicely inferred by the _____ lemma.

Select the correct option

○	addition
○	parenthesis
◉	node
○	division

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Question # 10 of 10 (Start time: 02:37:57 PM, 02 July 2025)

The _____ given by DFS allow us to determine whether the graph contains any cycles.

Select the correct option

◉	Time stamps
○	Order
○	BFS traversing
○	Topological sort

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Question # 3 of 10 (Start time: 02:35:49 PM, 02 July 2025)

In a digraph, the number of edges coming out of a vertex is called the _____ of that vertex.

Select the correct option

○	In-degree
○	Complete degree
◉	Out-degree
○	Node

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Question # 6 of 10 (Start time: 02:36:37 PM, 02 July 2025)

In digraph G=(V,E) ;G has cycle if and only if

Select the correct option

◉	The DFS forest has back edge
○	The DFS forest has forward edge.
○	BFS forest has forward edge
○	The DFS forest has both back and forward edge

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Question # 7 of 10 (Start time: 02:36:55 PM, 02 July 2025)

In computing the _____ components of a digraph, vertices of the digraph are partitioned into subsets.

Select the correct option

◉	strongly connected
○	best
○	weakly connected
○	worst

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Question # 5 of 10 (Start time: 02:36:14 PM, 02 July 2025)

Forward edge is:

Select the correct option

○	(u, v) where v is a proper ancesstor of u in the tree.
○	(u, v) where u is a proper ancesstor of v in the tree.
●	(u, v) where v is a proper descendent of u in the tree.
○	(u, v) where u is a proper descendent of v in the tree.

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Question # 2 of 10 (Start time: 02:35:37 PM, 02 July 2025)

The relationship between number of back edges and number of cycles in DFS is,

Select the correct option

○	Back edges are one quarter of cycles
◉	Both are equal
○	Back edges are half of cycles
○	There is no relationship between no. of edges and cycles